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Water Purifier Service Management System (WPSMS)

Software Requirements & Technical Architecture Document

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1. PROJECT OVERVIEW

1.1 Introduction

The Water Purifier Service Management System (WPSMS) is a centralized digital platform designed to manage the allocation, installation, servicing, diagnostics, and lifecycle tracking of water purifier devices across distributed operational networks.

The system connects administrators, distributors, local partners, and field technicians through standardized workflows and real-time information exchange.

1.2 Business Objectives

- Ensure end-to-end device traceability
- Digitize installation and service workflows
- Enable fraud-proof job completion
- Provide live operational visibility
- Maintain permanent service history
- Enable hardware diagnostics via Bluetooth

2. Operational Hierarchy



3. SYSTEM ARCHITECTURE

3.1 Components

- Web Admin Pane
- Technician Mobile Application
- Central API & Database
- Bluetooth Communication Layer
- Reporting & Analytics Engine

3.2 High Level Flow

Device creation → allocation → installation → active service → maintenance → parts-replacement → historical archive.

4. User Roles

Role	Description
Super Admin	Global governance & visibility
Distributor	Regional stock & assignment
Local Partner	Technician coordination
Technician	Installation & service execution

5. Admin Panel Workflow

Used by management teams for operational control.

5.1 Device Lifecycle

1. Device created with unique ID / QR.
2. Assigned from Super Admin → Distributor → Local Distributor.
3. Allocated to technician.
4. Installed at customer location.
5. All future actions recorded permanently

5.2 Client Benefits

- Warranty validation
- AMC enforcement
- Fraud prevention
- Instant service history
- Asset visibility

6. Installation Request Flow

1. Admin creates order.
2. Assigned to technician.
3. Technician notified in mobile app.
4. Site visit → installation → photo upload.
5. Device completed → device activated

7. Service / Complaint Flow

Requests can be:

- Manual
- Auto-generated

Assignment:

- Automatic
- Manual by manager

Technician must:

Accept → Travel → Repair → Upload proof → Close.

Admins monitor progress in real time.

8. Management Visibility Dashboard

Management can track:

- Pending vs completed jobs
- Technician productivity
- Regional performance
- Spare consumption
- Rejections

9. Technician Mobile Application

9.1 Login & Identity Tracking/Home Dashboard

System records:

- ✓ device identity
- ✓ login history
- ✓ notification token

Technician sees:

- Assigned jobs
- Daily metrics
- Alerts
- Service history access

10. Task Lifecycle (Engineer)

Assigned → Accepted → In Progress → Completed

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Rejected / Waiting

Rules

- Reject → reason mandatory
- Waiting → explanation mandatory
- Complete → photos mandatory

11. On-Site Service Procedure

- Technician must:
- Scan QR
- Verify device
- View history

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- Replace parts
 - Run Bluetooth diagnostics
 - Upload photos
 - Submit report

12. Photo Proof System

Up to 3 images:

- Before service
- After service
- Replaced component

13. Bluetooth Service Workflow

Technician App ↔ Purifier Controller.

13.1 Operational Goals

- Secure connection
- Identity validation
- Live telemetry
- Configuration
- Service reset

13.2. Communication Standard

All messages use JSON.

Mandatory fields:

- command
- timestamp
- device identity
- status (response)

13.3. Device Handshake Example

Protocol Rules

Every request must contain:

- **command**
- **timestamp**

Responses must contain:

- **timestamp**
- **status**
- **message**

Status Codes

Value	Meaning
1	Success
0	Failure

13.3.1 Device Connect Handshake

Purpose

Establish a secure session and confirm the controller is reachable.

Pre-Condition

Technician is physically near the purifier and Bluetooth is enabled.

Post-Condition

If successful, the session becomes active and further commands are allowed.

App → Device (Request)

```
{
  "command": "connect",
  "bluetooth_mac": "A4:C1:38:9F:21:7B",
  "timestamp": "2025-01-15T10:45:30Z"
}
```

Device → App (Success Response)

```
{
```

```
"bluetooth_mac": "A4:C1:38:9F:21:7B",
"timestamp": "2025-01-15T10:45:31Z",
"status": 1,
"message": "Connection successful"
}
```

Device → App (Failure Response)

```
{
  "bluetooth_mac": "A4:C1:38:9F:21:7B",
  "timestamp": "2025-01-15T10:45:31Z",
  "status": 0,
  "message": "Connection failed"
}
```

System Behaviour

If status = 1
→ session established
→ technician can read data / reset / configure.

If status = 0
→ technician must retry connection.

13.3.2. Reset Command

Purpose

Clear service counters, alarms, and refresh controller state after maintenance.

Pre-Condition

Active Bluetooth session must exist.

Post-Condition

New lifecycle begins from reset timestamp.

App → Device (Request)

```
{
  "command": "reset",
  "timestamp": "2025-01-15T10:50:00Z"
}
```

Device → App (Success Response)

```
{  
  "command": "reset",  
  "timestamp": "2025-01-15T10:50:05Z",  
  "status": 1,  
  "message": "Device reset completed"  
}
```

Device → App (Failure Response)

```
{  
  "command": "reset",  
  "timestamp": "2025-01-15T10:50:05Z",  
  "status": 0,  
  "message": "Reset failed"  
}
```

14. Conclusion

The WPSMS platform enables structured coordination between administrators, partners, and field engineers. Every stage — from request generation to closure — is digitally authenticated, measurable, and auditable.

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